Ayushmaan Puri

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Education

University of California, San Diego September 2024 - September 2026
B.S. CSE: Computer Engineering *GPA 3.40/4.0*

Orange Coast College August 2022 - May 2024
A.S. for Transfer: Computer Science, Mathematics, and Physics – Graduated with Honors *GPA 3.87/4.0*

Research Experience

Undergraduate Researcher | Junkyard Computing March 2026 – September 2026
Kastner Research Group – UC San Diego *La Jolla, CA*

Batteryless Operation and Thermal Testing *June 2026 – September 2026*

- Enabled batteryless operation of stripped-down Pixel Fold nodes by wiring regulated DC power to the mainboard and spoofing the battery ID and NTC thermistor signals, found by reverse-engineering the dual Samsung PMIC (s2mpg12/s2mpg13) boot sequence in the felix/gs201 kernel tree; this frees the USB-C port for networking and removes thousands of lithium cells from the rack.
- Ran per-cluster `stress-ng` workloads while logging thermal zones (`/sys/class/thermal/`), on-device ODPM power rails, and wall-side power from the bench PSU over its USB interface, to physically locate the Tensor G2 CPU/GPU clusters on the mainboard and map hotspots for cooling design and throttling prevention.
- Coordinating networking and deployment logistics to integrate the phone racks into UCSD's DSMLP data center for course autograding, working across the thermal/rack, teardown, and kernel teams and the data center staff.

Autograder Cluster *March 2026 – June 2026*

- Repurposed discarded smartphones as a dedicated autograding cluster to replace oversubscribed GPU pods shared across a class of 330 students.
- Deployed a 40-node k3s Kubernetes cluster on ARM64 Google Pixel Fold devices running Debian over USB Ethernet; configured each node as an isolated Linux execution environment for containerized grading jobs.
- Modeled real submission timing and runtime distributions from 8 programming assignments using Gaussian Mixture Models with BIC-selected component counts.
- Built a synthetic workload generator sampling from fitted GMMs to stress test cluster throughput and determine the minimum phone count needed to serve a class without queuing delays.

Undergraduate Researcher | UnifiedSplitting June 2025 – February 2026
PatLab & UC Scholars Research Program – UC San Diego *La Jolla, CA*

- Built heterogeneous CPU-GPU pipelines for layer-split CNN inference on NVIDIA Jetson unified-memory hardware.
- Profiled per-layer CPU-GPU speedup ratios across full network architectures; identified a 10x speedup gap between GPU inference of convolutional layers (300–500x) and fully connected layers (30–40x) to determine optimal split points.
- Tested the approach across five architectures (ResNet-18/50, MobileNetV2, EfficientNet-B2, ViT-Tiny) and six simulated edge hardware profiles; compared them to GPU-only baselines.

Summer Research Intern | Active Thermography for Defect Detection June 2024 – Sep. 2024
SeNSE – Indian Institute of Technology, Delhi *New Delhi, India*

- Developed a per-pixel signal processing pipeline in Python (OpenCV, NumPy, SciPy) to detect subsurface defects in Carbon and Glass Fibre-Reinforced Polymer (CFRP/GFRP) composites using active thermography.
- Implemented linear detrending, full temporal cross-correlation against a sound reference pixel, main lobe extraction, and FFT phase computation across three experimental datasets (LFM, Active Pulsed, and Flux excitation).
- Demonstrated that the FFT phase of the correlation main lobe produces qualitatively distinct signatures for defective vs. healthy material regions, establishing a new discriminating observable for non-destructive evaluation. Recorded results and findings in a research paper.

Summer Research Intern | Lunar Rover Modeling June 2023 – Aug. 2023
NASA California Space Grant Consortium *Costa Mesa, CA*

- Designed and built a model lunar lava tube exploration rover using a rocker-bogie chassis, solar-powered with lithium-polymer battery storage.
- Developed Arduino firmware integrating UV, Hall effect magnetic, and gravimeter sensors to detect water-ice signatures, iron-rich rock, and subsurface density anomalies.
- Implemented on-board SD card data logging for sensor telemetry and an object avoidance drive system for autonomous navigation over loose regolith.

Publications and Talks

UnifiedSplitting: CPU–GPU Split Inference on Edge Devices | *Oral presentations* 2025

- Southern California Conference for Undergraduate Research (SCCUR), CSU Channel Islands, Nov. 2025, with Fahad Alkhazam and Parth Mehta. [Slides] [Abstract]
- UC San Diego Summer Research Conference (SRC), Aug. 2025. [Abstract]

Awards and Honors

CRA Outstanding Undergraduate Researcher Honorable Mention | *Computing Research Association* Jan. 2026

UC Scholar ($5000) | *UC San Diego* May 2025

George Ciarlo Memorial Scholarship ($1000) | *Orange Coast College* Apr. 2024

Teaching, Mentorship, and Service

Research Mentor – Engineers for Exploration (E4E) June 2026 – Aug. 2026
Matthew Henson Fellowship – UC San Diego *La Jolla, CA*

- Mentored visiting undergraduate researchers from Howard University on the Junkyard smartphone cluster project.
- Guided students through SoC stress-testing on repurposed Pixel devices and custom server rack design for logic boards.

Conference Moderator August 2026
UC San Diego Summer Research Conference 2026 *La Jolla, CA*

- Moderated two sessions with 4 presenters each. Computer Science for the Public Good and Research in Electrical, Computer, and Mechanical Engineering

STEM Peer Mentor Aug. 2023 – May 2024
OCC STEM Academy / MESA Program *Costa Mesa, CA*

- Mentored 17 first-generation students, providing weekly 1-on-1 guidance, email check-ins, and long-term academic support.
- Organized and led workshops on career development, internships, and transfer preparation.
- Co-managed the STEM Center with 6 peer mentors, coordinating student outreach and resource distribution.

Club President Aug. 2023 – May 2024
OCC STEM Club *Costa Mesa, CA*

- Hosted weekly workshops and projects for 20+ members across all STEM disciplines, fostering an inclusive community for students to explore diverse scientific interests.
- Organized volunteering missions to foster interest in STEM fields among children. Led a group of 10 volunteers to teach young students about physics and biology at OCC's annual science night.

Volunteer Teaching Assistant July 2023 – Feb. 2024
Microsoft TEALS Program *Houston, TX (Remote)*

- Co-taught CS Discoveries curriculum twice weekly to a class of 26 middle school students, integrating lessons on programming logic and problem-solving.
- Designed a sustainable CS education framework, enabling the program to continue beyond the volunteer period.

Technical Skills

Languages: C/C++, Python, SystemVerilog, MIPS Assembly, MATLAB, JavaScript
Parallel & ML: CUDA, OpenCL, PyTorch, scikit-learn, NumPy/SciPy, OpenCV
Hardware & EDA: Cadence Virtuoso, HSPICE, NVIDIA Jetson, ESP32, ARM64

Projects

Tiled Matrix Multiplication Visualizer | *HTML, JS, Parallel Programming* Feb. 2026

- Built an interactive visualizer, deployed at aypuri.com/tiledmatmultvis, to demonstrate tiled matrix multiplication, data reuse, and memory access patterns.

ESP32 Bluetooth Speaker | *Microcontrollers, Circuit Design, C++* Apr. 2025

- Built a Bluetooth speaker with a display showing title, artist, and album, powered by an ESP32 microcontroller.

DSA Visualizer | *TypeScript, Web Development* Apr. 2024

- Co-created a web application, deployed at animated-octo-spork.vercel.app, with a TypeScript back-end to help students visualize core data structures (linked lists, trees, graphs, stacks, queues).